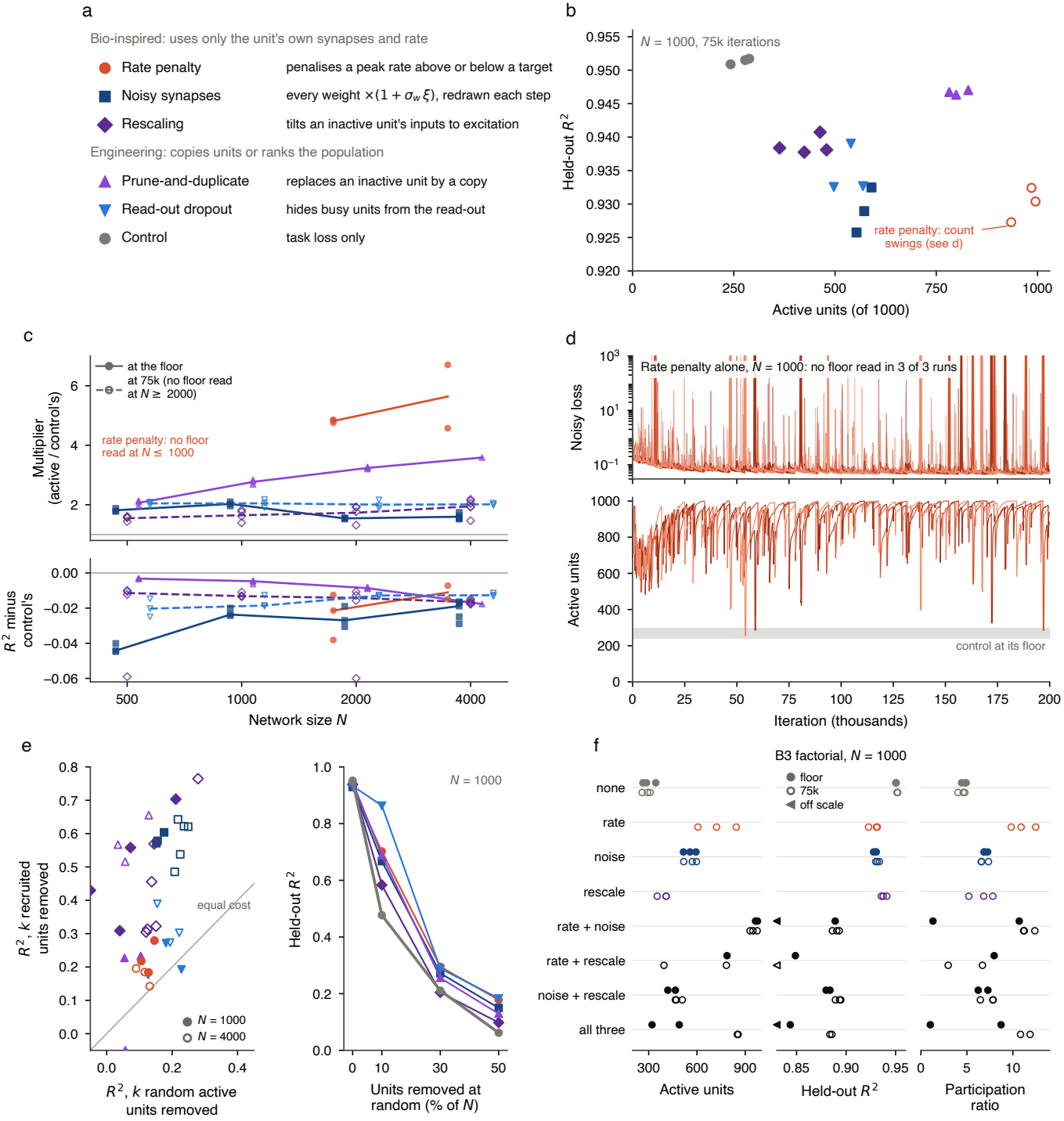

a
Bio-inspired: uses only the unit's own synapses and rate
Rate penalty
penalises a peak rate above or below a target
Noisy synapses
every weight ×(1 + σw ξ), redrawn each step
Rescaling
tilts an inactive unit's inputs to excitation
Engineering: copies units or ranks the population
Prune-and-duplicate
replaces an inactive unit by a copy
Read-out dropout
hides busy units from the read-out
Control
task loss only
b
N = 1000, 75k iterations
Held-out R2
Active units (of 1000)
rate penalty: count swings (see d)
c
at the floor
at 75k (no floor read at N ≥ 2000)
rate penalty: no floor read at N ≤ 1000
Multiplier (active / control's)
R2 minus control's
Network size N
d
Rate penalty alone, N = 1000: no floor read in 3 of 3 runs
Noisy loss
Active units
control at its floor
Iteration (thousands)
e
equal cost
N = 1000
N = 4000
R2, k recruited units removed
R2, k random active units removed
N = 1000
Held-out R2
Units removed at random (% of N)
f
B3 factorial, N = 1000
floor
75k
off scale
none
rate
noise
rescale
rate + noise
rate + rescale
noise + rescale
all three
Active units
Held-out R2
Participation ratio